

TrES-3b

R-0061

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_251002 · created 2026-07-15T14:45:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460950.6488 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.027
Transit depth (Rp/Rs)^2	3.65 +/- 1.03 [%]
Orbital Inclination (inc)	81.6 +/- 1.48
Airmass coefficient 1 (a1)	1.029 +/- 0.014
Airmass coefficient 2 (a2)	-0.02 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	#3 - [158, 421]
Optimal Aperture	3.62
Optimal Annulus	12.51
Transit Duration (day)	0.053 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.027 vs 0.16309 (deviation 1.03σ)
Duration fit vs published	0.053 d vs 0.0567 d (deviation 0.22σ)
Mid-time O–C	-2.4 min
Depth SNR	7.1
Residual scatter	1.38 %

Light curve

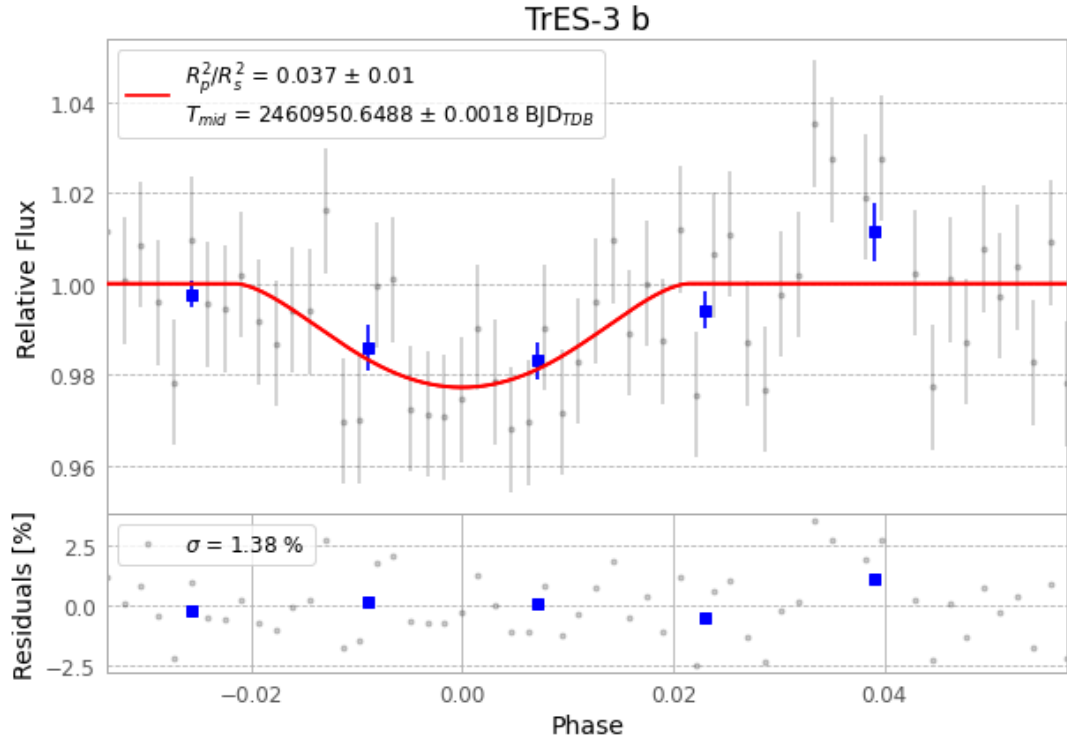


Figure 1 — FinalLightCurve_TrES-3 b_2025-10-01.png (SHA-256 388b3bc44c278c0e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [228, 335], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3169beb5ec0c21a1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

58 FITS frames (38 MB), TRES-3251002023015.FITS → TRES-3251002052115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-251002.log (7adb6d524451...), FinalLightCurve_TrES-3 b_2025-10-01.png (388b3bc44c27...), FinalLightCurve_TrES-3 b_2025-10-01.csv (9e9c14baee0d...)

Compute resources

Wall time 135.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.